

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0304	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	<i>Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	4	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	6	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	4	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	4	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	2	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

S.No	Register No.	Student Name	Question
1	192224283	PEDDAGANI RUTHISH	Design a concept map that evaluates the classes of mobile computing devices. • Relate smartphones, tablets, laptops, wearables and IoT devices to their processing power, battery, display and connectivity. • Assess which device class is best suited to continuous mobile computing. • Justify the linking phrases used between each device class and its constraints.
2	192311008	THALAMATI AASRITHA	Construct a concept map that evaluates context-awareness in mobile computing. • Relate location context, time context, user activity, device state and network conditions to adaptive services. • Assess which type of context adds the most value to mobile services. • Support your evaluation with meaningful concept relationships.
3	192311154	A KAVYA SREE	Develop a concept map that evaluates disconnected operation in mobile computing. • Relate hoarding, emulation (local operation), reintegration and conflict resolution. • Assess which phase is most critical for data consistency after reconnection. • Justify your evaluation through logically ordered concept links.
4	192311182	SRIPATHI MARAM	Design a concept map that evaluates the role of middleware in mobile computing. • Relate proxies, caching, prefetching, data compression and application adaptation. • Determine which middleware technique best hides network variability from the user. • Support your conclusion using appropriate linking phrases.
5	192311244	GAYAM RAHUL	Construct a concept map that evaluates the shift from large-zone radio systems to the cellular concept. • Relate single high-power transmitters, small low-power cells, frequency reuse and subscriber capacity. • Assess why the cellular concept solved the spectrum shortage problem. • Justify your evaluation with clear cause-and-effect relationships.
6	192311274	SURAPAREDDY REVANTH	Develop a concept map that evaluates the choice of frequency bands for cellular deployment. • Relate low bands (700/900 MHz), mid bands (1800/2100 MHz) and high bands (3.5 GHz and above) to coverage, penetration and capacity. • Recommend the best band mix for a region with both rural and dense urban areas. • Justify your recommendation through well-defined concept links.

7	192311287	MOPURI GOWTHAMI	Design a concept map that evaluates the use of macro, micro, pico and femto cells. • Relate cell size, transmit power, user density, indoor coverage and deployment cost. • Assess which cell type is most suitable for indoor hotspots such as malls and offices. • Support your evaluation with meaningful comparative relationships.
8	192324037	MANIVANNAN AISHWARYA	Construct a concept map that evaluates antenna techniques in cellular systems. • Relate directional antennas, antenna downtilt, space diversity and polarisation diversity to interference and fading. • Determine which technique gives the greatest improvement in link quality. • Justify the relationships shown in your concept map.
9	192324062	SANIA HERBERT	Develop a concept map that evaluates the near-far problem and uplink power control. • Relate distance from the base station, received power imbalance, open-loop and closed-loop power control and battery life. • Assess why power control is more critical in CDMA systems than in TDMA systems. • Support your evaluation using appropriate linking phrases.
10	192324289	RAKINA S	Design a concept map that evaluates interference-limited and noise-limited cellular systems. • Relate thermal noise, co-channel interference, cell size, traffic load and capacity. • Assess under which conditions a system moves from noise-limited to interference-limited. • Justify your evaluation through interconnected concepts.
11	192324306	CHODAM NISHA	Construct a concept map that evaluates timing advance and guard periods in TDMA access. • Relate propagation delay, mobile distance, timing advance, guard period and maximum cell radius. • Assess how timing advance prevents overlap between time slots of different users. • Justify your conclusions using meaningful cause-and-effect links.
12	192371005	VENKATA JAYANI B	Develop a concept map that evaluates the mapping of logical channels onto physical channels. • Relate traffic and control logical channels, time slots, carrier frequencies and multiframe structures. • Assess how this mapping makes efficient use of radio resources. • Support your evaluation with meaningful hierarchical relationships.
13	192371030	SANTHIYA K M	Design a concept map that evaluates how a mobile station synchronises with a cell. • Relate the frequency correction channel, synchronisation channel, broadcast control channel and system information. • Determine which step is most essential before a mobile can access the network. • Justify the order of concepts shown in your concept map.

14	192372002	SOMA GEETHA MOULIKA	Construct a concept map that evaluates frequency hopping in cellular systems. • Relate slow frequency hopping, hopping sequences, interference diversity and frequency diversity. • Assess how frequency hopping improves call quality for slow-moving users. • Support your evaluation using appropriate concept links.
15	192372004	GAGGUTURI ALTHAF	Develop a concept map that evaluates the functions of the Mobile Switching Centre. • Relate call control, switching, mobility support, call detail record generation and interworking with other networks. • Assess which MSC function is most critical for service continuity. • Justify your evaluation through logical concept organisation.
16	192372015	PEDABALLE MAL REDDY	Design a concept map that evaluates the components of a cell site. • Relate antennas, feeder cables, the BTS cabinet, power backup (battery/generator) and backhaul equipment. • Determine which component failure has the greatest impact on site availability. • Justify the relationships established in your concept map.
17	192372021	JAGARLAMUDI SUPREETH KUMAR	Construct a concept map that evaluates backhaul options for cellular base stations. • Compare optical fibre, microwave radio, satellite and copper links in terms of capacity, cost, latency and deployment time. • Recommend the most suitable backhaul for urban and remote sites. • Support your recommendation with well-defined conceptual relationships.
18	192372028	BANAGANAPALLI MUNNA	Develop a concept map that evaluates the mobile network protocol stack against the OSI reference model. • Relate the physical, data link and network layers of the radio interface to the corresponding OSI layers. • Assess where the mobile protocol stack differs from OSI and why. • Justify your evaluation using meaningful hierarchical links.
19	192372051	BANDILI KARTHIK RATAN	Design a concept map that evaluates the sublayers of the cellular network layer. • Relate Radio Resource management, Mobility Management and Connection Management to their procedures. • Assess which sublayer has the largest effect on call continuity. • Support your conclusions through appropriate concept relationships.
20	192372056	HASTHAVARAM SANJAY REDDY	Construct a concept map that evaluates error control on the radio link. • Relate forward error correction, interleaving, ARQ retransmission and burst errors. • Determine which technique best protects speech and which best protects data. • Justify your evaluation with meaningful linking phrases.
21	192372089	SIRIYALA SRIKANTH	Develop a concept map that evaluates the separation of the control plane and the user plane in the core network. • Relate signalling functions, user data forwarding, scalability and independent upgrades. • Assess the benefits of separation for handling growing data traffic. • Support your evaluation using appropriate concept links.

22	192372168	MOTHUKURI BHARATH	Design a concept map that evaluates reliability and redundancy in the mobile core network. • Relate MSC pooling, geo-redundant subscriber databases, load sharing and automatic failover. • Assess which mechanism most reduces service outage for subscribers. • Justify your evaluation through interconnected concepts.
23	192372226	Pittam Parthasaradhi Reddy	Construct a concept map that evaluates how a PLMN interconnects with other networks. • Relate the SS7 signalling network, signal transfer points, interconnect gateways, the PSTN, other PLMNs and the internet. • Assess the importance of the signalling network in interconnection. • Justify the relationships included in your concept map.
24	192372264	SHAIK SAMEENA	Develop a concept map that evaluates infrastructure sharing between mobile operators. • Relate passive sharing (towers, power), active RAN sharing, shared spectrum and cost savings. • Assess the trade-off between cost reduction and operator control. • Support your evaluation with meaningful concept relationships.
25	192372291	SHAIK ARSHIYA	Design a concept map that evaluates the roles of Mobile Network Operators and Mobile Virtual Network Operators in a PLMN. • Relate spectrum ownership, radio and core infrastructure, subscriber management, branding and wholesale agreements. • Assess how MVNOs increase competition without owning a radio network. • Justify your evaluation using appropriate linking phrases.
26	192372346	BIJJA SAI MADHAVA	Construct a concept map that evaluates mobile number portability. • Relate the MSISDN, porting database, donor and recipient networks, HLR updates and call routing. • Assess the effect of number portability on call routing efficiency. • Justify your conclusions through logical concept organisation.
27	192373003	CHITTABATTINA MRUDHULA	Develop a concept map that evaluates emergency call handling in cellular networks. • Relate emergency numbers, limited-service state, calls without a SIM, call priority and caller location. • Assess which feature most improves emergency response. • Support your evaluation through meaningful concept links.
28	192373032	BOLLIGORLA HEVANTH KUMAR	Design a concept map that evaluates user mobility models used in network design. • Relate the random walk model, fluid-flow model, cell residence time and boundary crossing rate. • Assess how the choice of mobility model affects location management design. • Justify the relationships shown in your concept map.
29	192421012	SUDHARSAN S	Construct a concept map that evaluates location update strategies. • Compare always-update, never-update, time-based, movement-based and distance-based updating. • Determine which strategy gives the best balance of signalling cost and accuracy. • Support your evaluation using appropriate linking phrases.

30	192421161	ABINAYA R	Develop a concept map that evaluates paging strategies in cellular networks. • Compare blanket paging, sequential paging and selective (probability-based) paging. • Assess the trade-off between paging delay and paging signalling load. • Justify your evaluation through well-defined concept relationships.
31	192472195	BANTROTHULA CHARAN KUMAR REDDY	Design a concept map that evaluates inter-system and vertical handoff. • Relate handoff between GSM, UMTS and LTE, and handoff between Wi-Fi and cellular, to session continuity and signalling. • Assess the main challenges of handing over between different technologies. • Support your evaluation with meaningful interconnected concepts.
32	192472213	KUPPAM SHALINI	Construct a concept map that evaluates fraud risks in roaming. • Relate SIM cloning, delayed roaming records, high-usage fraud, fraud detection and near-real-time data exchange. • Assess which control most reduces roaming fraud losses. • Justify your conclusions using appropriate concept links.
33	192512082	Imayavaramban D	Develop a concept map that evaluates data roaming architectures. • Compare home routing and local breakout in terms of latency, cost, service control and lawful interception. • Recommend the better option for latency-sensitive services. • Justify your recommendation through logical concept relationships.
34	192521040	Sudharsan R	Design a concept map that evaluates Quality of Service classes for mobile services. • Relate conversational, streaming, interactive and background classes to delay, jitter, throughput and error tolerance. • Assess which class places the strictest demands on the network. • Justify the hierarchy established in your concept map.
35	192521108	Bhuvanesh C	Construct a concept map that evaluates cell breathing in CDMA-based systems. • Relate traffic load, interference level, cell coverage area and power control. • Assess how cell breathing affects coverage at busy hours. • Support your evaluation using meaningful cause-and-effect links.
36	192521272	Angel Deepika M	Develop a concept map that evaluates the drive test and network optimisation cycle. • Relate measurement collection, KPI analysis, problem identification, parameter tuning and verification. • Determine which stage contributes most to improving network performance. • Justify the sequence of concepts shown in your concept map.
37	192521455	Dinesh M	Design a concept map that evaluates the root causes of dropped calls. • Relate coverage holes, interference, handoff failures, congestion and equipment faults. • Rank the causes by their impact in a typical urban network. • Support your ranking with appropriate concept relationships.

38	192524242	Yogavan Nidhi Y	Construct a concept map that evaluates traffic variation and busy-hour dimensioning. • Relate busy-hour traffic, daily traffic profiles, hotspots, non-uniform user distribution and channel dimensioning. • Assess why networks are dimensioned for the busy hour rather than average traffic. • Justify your evaluation through meaningful concept links.
39	1912260451	Yuvasri M	Develop a concept map that evaluates the provisioning of a new mobile subscriber. • Relate SIM issue, IMSI and Ki allocation, HLR and AuC entries, MSISDN assignment and service activation. • Determine which provisioning step is most critical for secure first registration. • Support your evaluation using a logically ordered concept structure.
40	1925260038	Kruthikaran	Design a concept map that evaluates charging and billing operations in a cellular network. • Relate call detail records, mediation, rating, postpaid billing and real-time prepaid charging. • Assess which component is most important for accurate and timely billing. • Justify your evaluation through well-defined conceptual relationships.